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INSTITUT UNIVERSITAIRE DES SCIENCES (IUS)

Faculté des sciences et des technologies (FST)

Rapport laboratoire TD N° 6 – Réseaux I

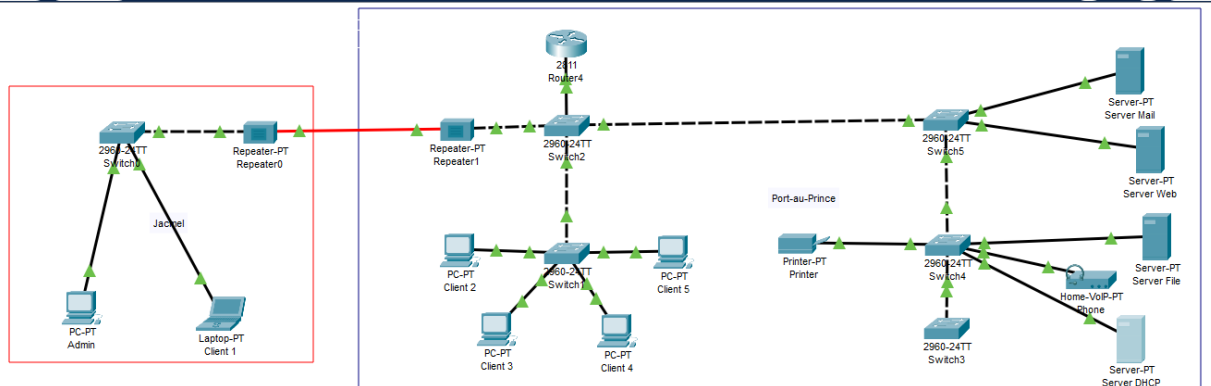
Préparé par : Brian ANTOINE

Présenté à: M. Ismaël SAINT AMOUR

Niveau : L3

Date: 06/12/2025

- 1) Reproduisons cette topologie en configurant les services DHCP afin d'attribuer automatiquement les adresses IP aux dispositifs du réseau :



➤ Configuration du serveur DHCP

Server DHCP

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 192.168.1.1

DNS Server: 0.0.0.0

Start IP Address: 192 168 1 3

Subnet Mask: 255 255 255 0

Maximum Number of Users: 30

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168.1.1	0.0.0.0	192.168.1.3	255.255.255.0	30	0.0.0.0	0.0.0.0

☐ Top

➤ Ajout automatique des ip address ipv4 provenant serveur DHCP

Admin

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.1.4

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::201:64FF:FEA0:28AA

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

☐ Top

Client 1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address 192.168.1.5

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::2D0:BAFF:FEC8:5753

Default Gateway

DNS Server

802.1X

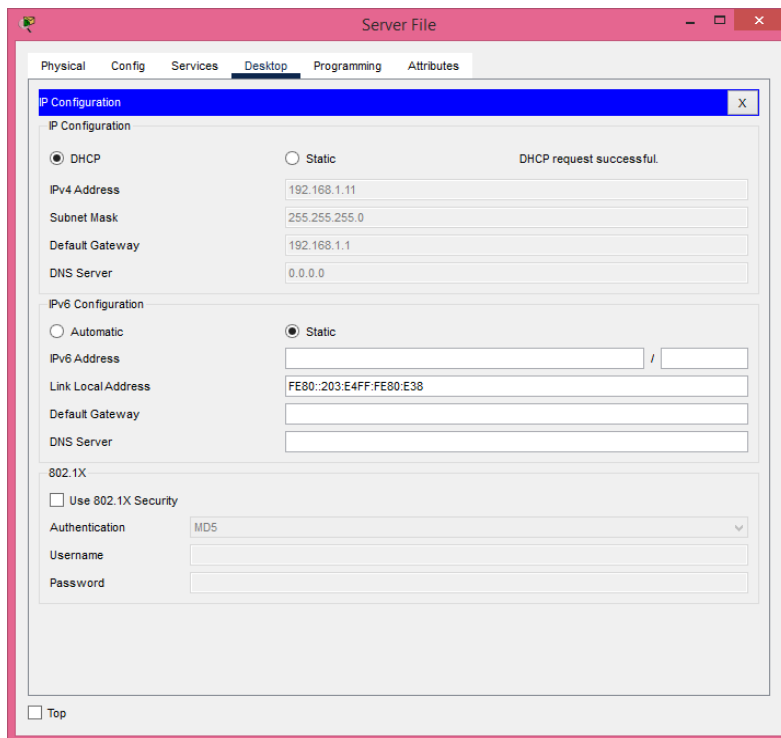
☐ Use 802.1X Security

Authentication MD5

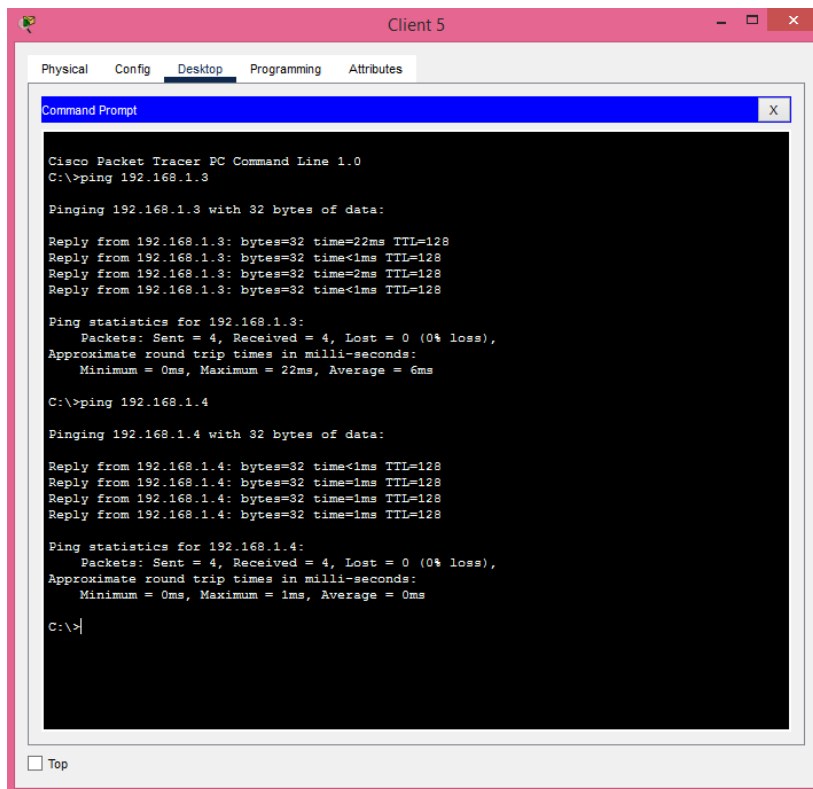
Username

Password

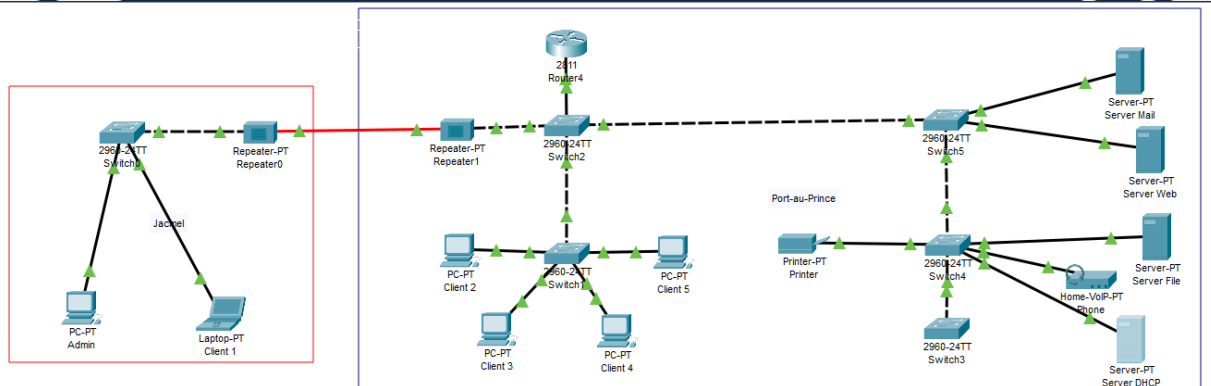
☐ Top



➤ Testons la connectivité avec ping



2. Reproduisons cette topologie en configurant les services DHCPv6 afin d'attribuer automatiquement les adresses IP aux dispositifs du réseau.



➤ Configuration du serveur DHCPv6

Server DHCP

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP
- DHCPv6**
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCPv6

Interface: FastEthernet0 Service: ☒ On ☐ Off

DHCPv6 Pool: 1

DHCPv6 Pool:

Pool List: 1 [Create Pool] [Remove Pool]

DNS Server: [] Domain Name: []

IPv6 Address Prefix

Prefix	Prefix Length	Valid Lifetime	Preferred Lifetime
[Create] [Edit] [Remove]			

IPv6 Prefix-Delegation

Prefix	DUID	Local Pool	Valid Lifetime	Preferred Lifetime
[Create] [Edit] [Remove]				

IPv6 Local Pool

Pool Name	Prefix	Prefix Length
[Create] [Edit] [Remove]		

☐ Top

➤ Ajout automatique des ip address ipv6 par le serveur DHCPv6

Admin

Physical Config Desktop Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address: 192.168.1.4

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 0.0.0.0

IPv6 Configuration

☒ Automatic ☐ Static ipv6 request successful.

IPv6 Address: 2001:DB8:1:0:F1E9:9D6D:C8C1:C8D1 / 64

Link Local Address: FE80::201:64FF:FEA0:28AA

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top

Client 1

Physical Config Desktop Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address: 192.168.1.5

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 0.0.0.0

IPv6 Configuration

☒ Automatic ☐ Static ipv6 request successful.

IPv6 Address: 2001:DB8:1:0:DE9E:9E3:A566:3457 / 64

Link Local Address: FE80::2D0:BAFF:FEC8:5753

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

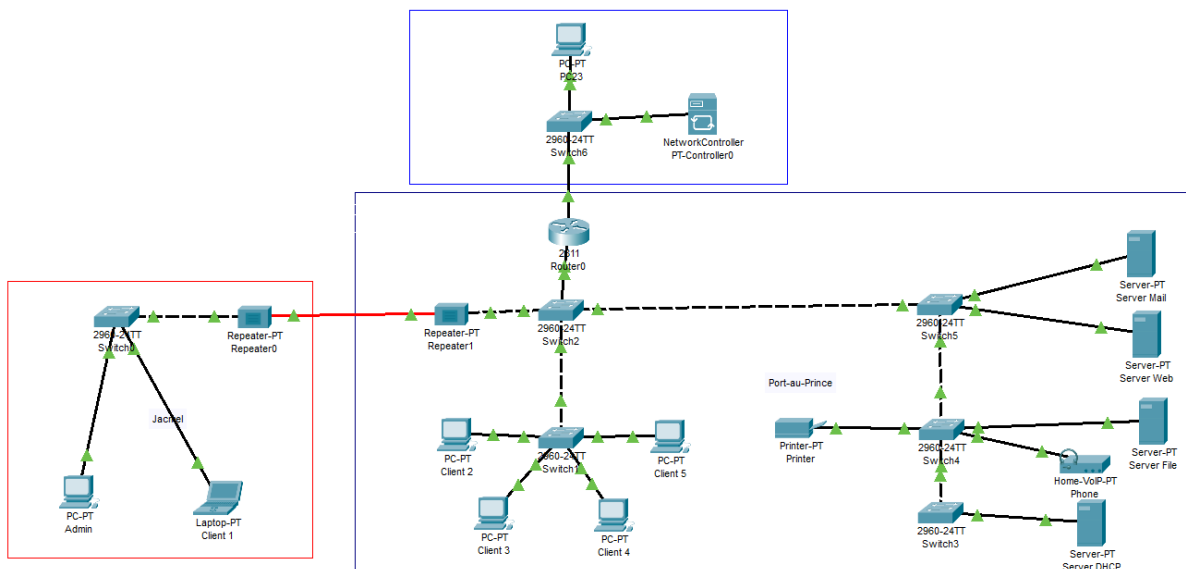
Authentication: MD5

Username:

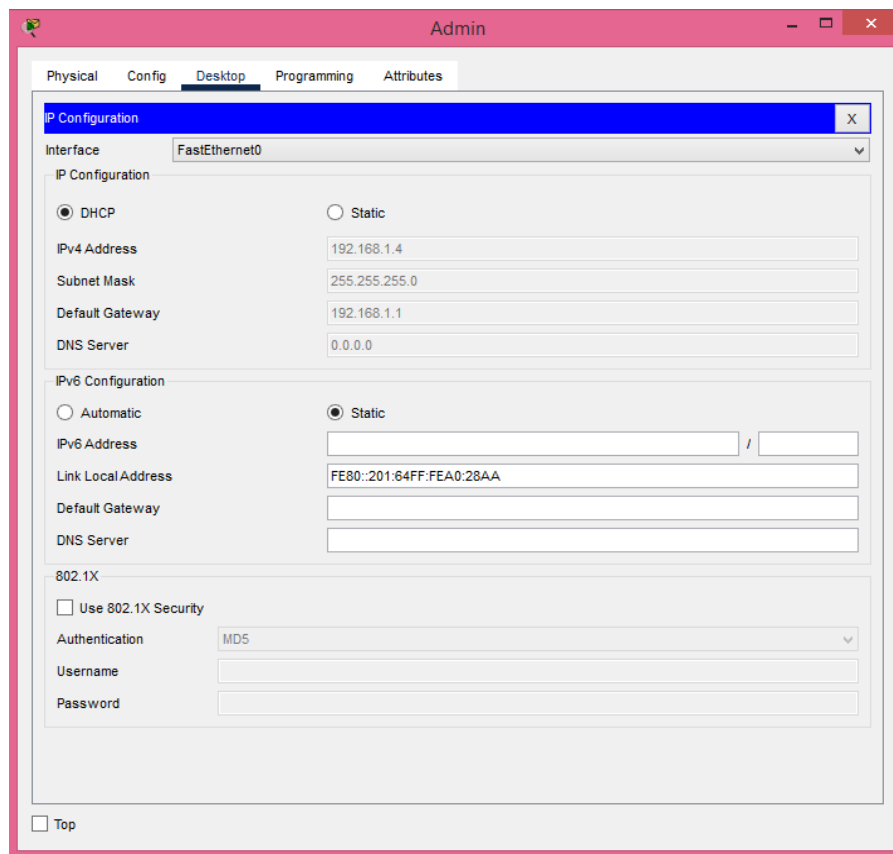
Password:

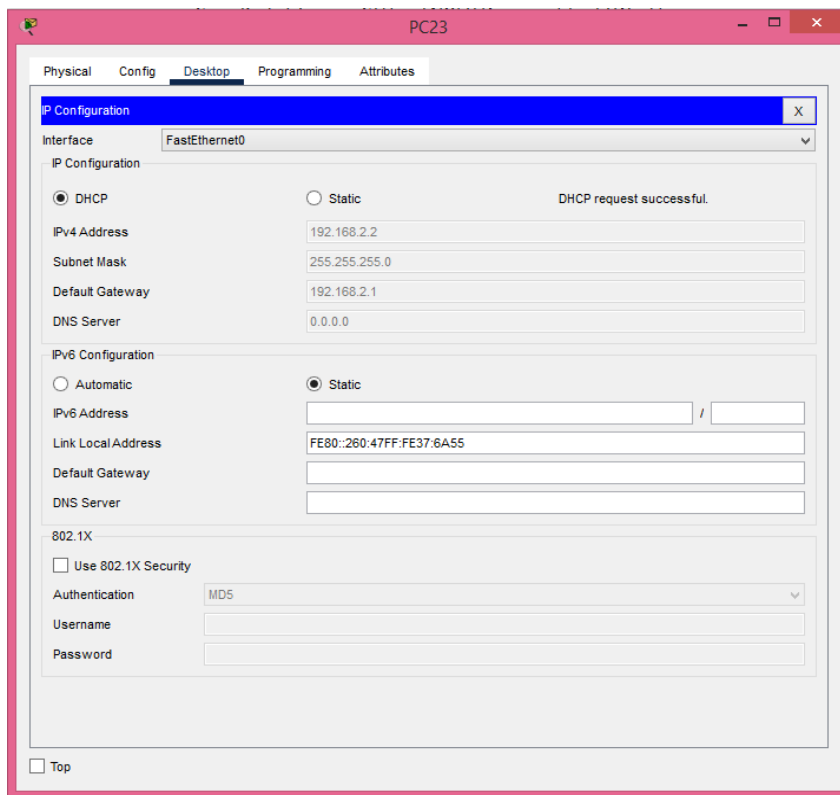
☐ Top

3. Reproduisons cette topologie en configurant les services DHCP, en utilisant le routeur comme serveur DHCP afin d'attribuer automatiquement les adresses IP aux différents hôtes du réseau.

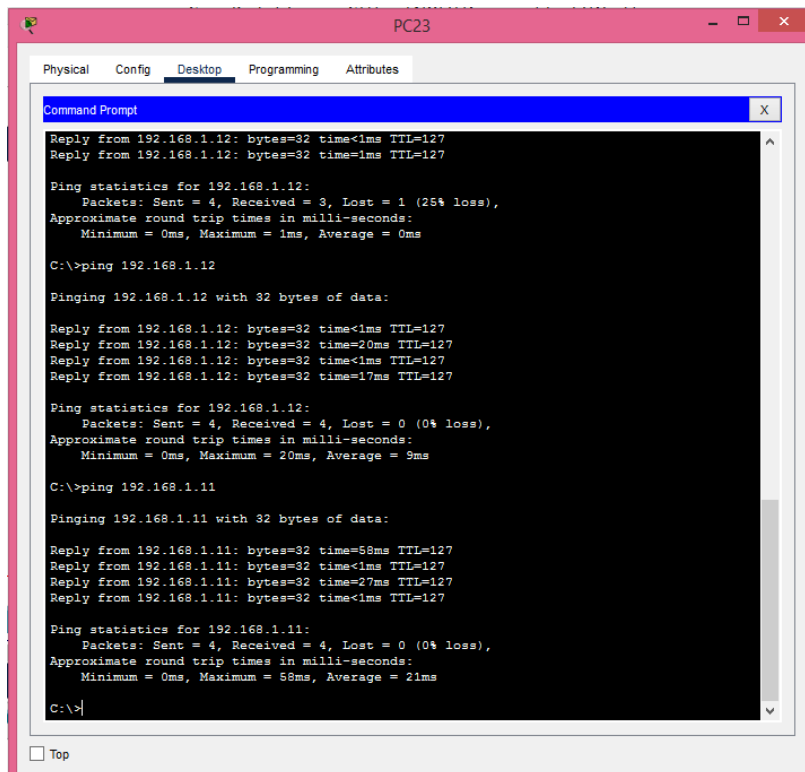


➤ **Ajout automatique des ip address ipv4 par le router utilisé comme serveur DHCP.**

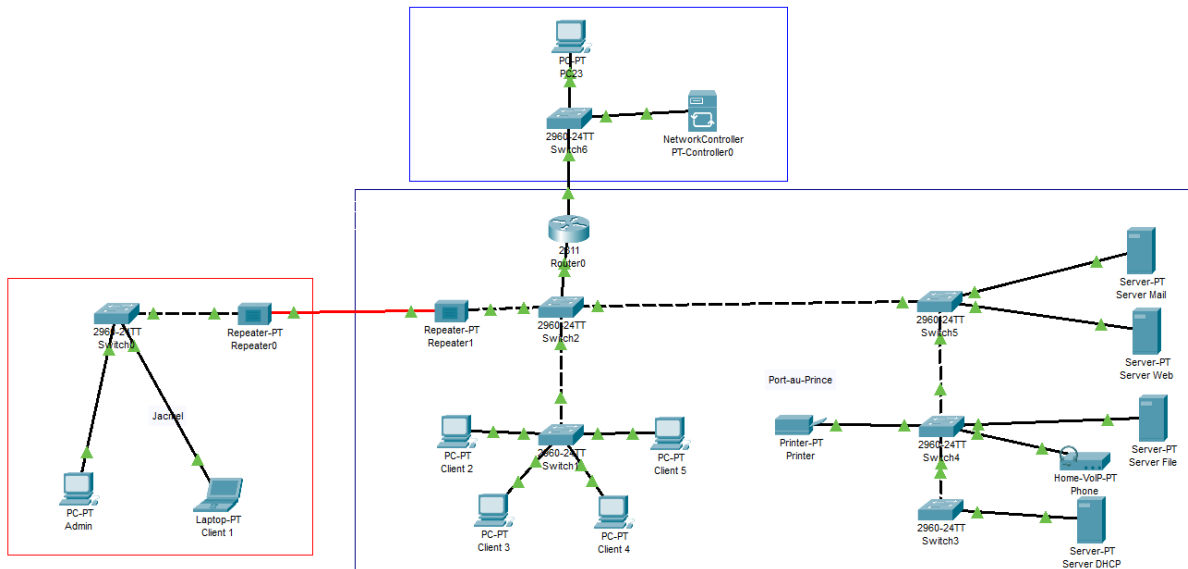




➤ *Testons la connectivité avec ping*



4. Reproduisons cette topologie en configurant les services DHCPv6, en utilisant le routeur comme serveur DHCP afin d'attribuer automatiquement les adresses IP aux différents hôtes du réseau.



- Ajout automatique des ip address ipv6 par le router utilisé comme serveur DHCPv6.

Client 3

Physical Config Desktop Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static DHCP request successful.

IPv4 Address: 192.168.1.11

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 0.0.0.0

IPv6 Configuration

☒ Automatic ☐ Static ipv6 request successful.

IPv6 Address: 2001:DB8:1:0:2A7F:55D4:F247:9ECA / 64

Link Local Address: FE80::2E0:F9FF:FECD:5690

Default Gateway: FE80::2D0:FFFF:FE15:1D01

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top

Client 2

Physical Config Desktop Programming Attributes

IP Configuration

Interface: FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address: 192.168.1.9

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

DNS Server: 0.0.0.0

IPv6 Configuration

☒ Automatic ☐ Static

IPv6 Address: 2001:DB8:1:0:1A7A:A85B:D4B0:D4B0 / 64

Link Local Address: FE80::290:2BFF:FE44:B2ED

Default Gateway: FE80::2D0:FFFF:FE15:1D01

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: MD5

Username:

Password:

☐ Top

Server Mail

Physical Config Services Desktop Programming Attributes

IP Configuration X

IP Configuration

☒ DHCP
 ☐ Static
 DHCP request successful.

IPv4 Address	192.168.1.13
Subnet Mask	255.255.255.0
Default Gateway	192.168.1.1
DNS Server	0.0.0.0

IPv6 Configuration

☒ Automatic
 ☐ Static
 Ipv6 request successful.

IPv6 Address	2001:DB8:1:0:82C2:BD26:599A:E88B / 64
Link Local Address	FE80::20C:CFFF:FE51:BB5D
Default Gateway	FE80::2D0:FFFF:FE15:1D01
DNS Server	

802.1X

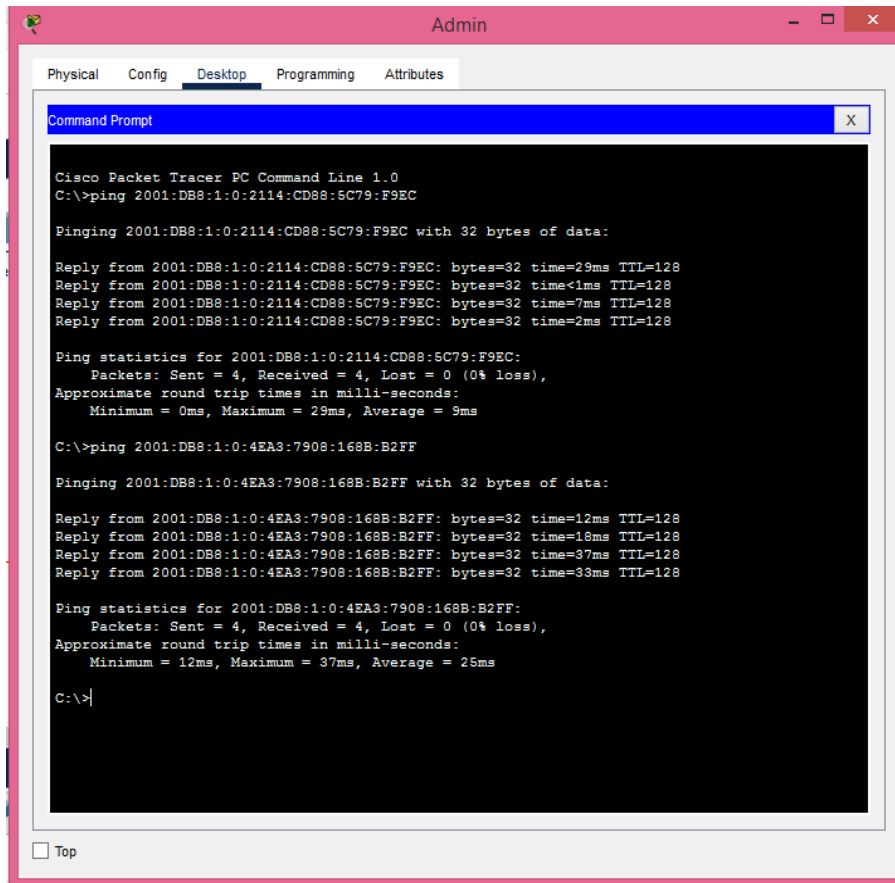
☐ Use 802.1X Security

Authentication MD5 ▼

Username	
Password	

☐ Top

➤ Testons la connectivité avec ping



The screenshot shows a Cisco Packet Tracer PC Command Line window titled "Admin". The window has tabs for "Physical", "Config", "Desktop", "Programming", and "Attributes", with "Desktop" selected. Inside the window is a "Command Prompt" window with a blue title bar and a close button. The command prompt displays the output of two ping commands. The first command is "C:\>ping 2001:DB8:1:0:2114:CD88:5C79:F9EC", which results in four successful replies with varying times and a statistics summary showing 0% loss. The second command is "C:\>ping 2001:DB8:1:0:4EA3:7908:168B:B2FF", which also results in four successful replies with varying times and a statistics summary showing 0% loss. The command prompt ends with "C:\>|".

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2001:DB8:1:0:2114:CD88:5C79:F9EC

Pinging 2001:DB8:1:0:2114:CD88:5C79:F9EC with 32 bytes of data:

Reply from 2001:DB8:1:0:2114:CD88:5C79:F9EC: bytes=32 time=29ms TTL=128
Reply from 2001:DB8:1:0:2114:CD88:5C79:F9EC: bytes=32 time<1ms TTL=128
Reply from 2001:DB8:1:0:2114:CD88:5C79:F9EC: bytes=32 time=7ms TTL=128
Reply from 2001:DB8:1:0:2114:CD88:5C79:F9EC: bytes=32 time=2ms TTL=128

Ping statistics for 2001:DB8:1:0:2114:CD88:5C79:F9EC:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 29ms, Average = 9ms

C:\>ping 2001:DB8:1:0:4EA3:7908:168B:B2FF

Pinging 2001:DB8:1:0:4EA3:7908:168B:B2FF with 32 bytes of data:

Reply from 2001:DB8:1:0:4EA3:7908:168B:B2FF: bytes=32 time=12ms TTL=128
Reply from 2001:DB8:1:0:4EA3:7908:168B:B2FF: bytes=32 time=18ms TTL=128
Reply from 2001:DB8:1:0:4EA3:7908:168B:B2FF: bytes=32 time=37ms TTL=128
Reply from 2001:DB8:1:0:4EA3:7908:168B:B2FF: bytes=32 time=33ms TTL=128

Ping statistics for 2001:DB8:1:0:4EA3:7908:168B:B2FF:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 37ms, Average = 25ms

C:\>|
```

☐ Top

